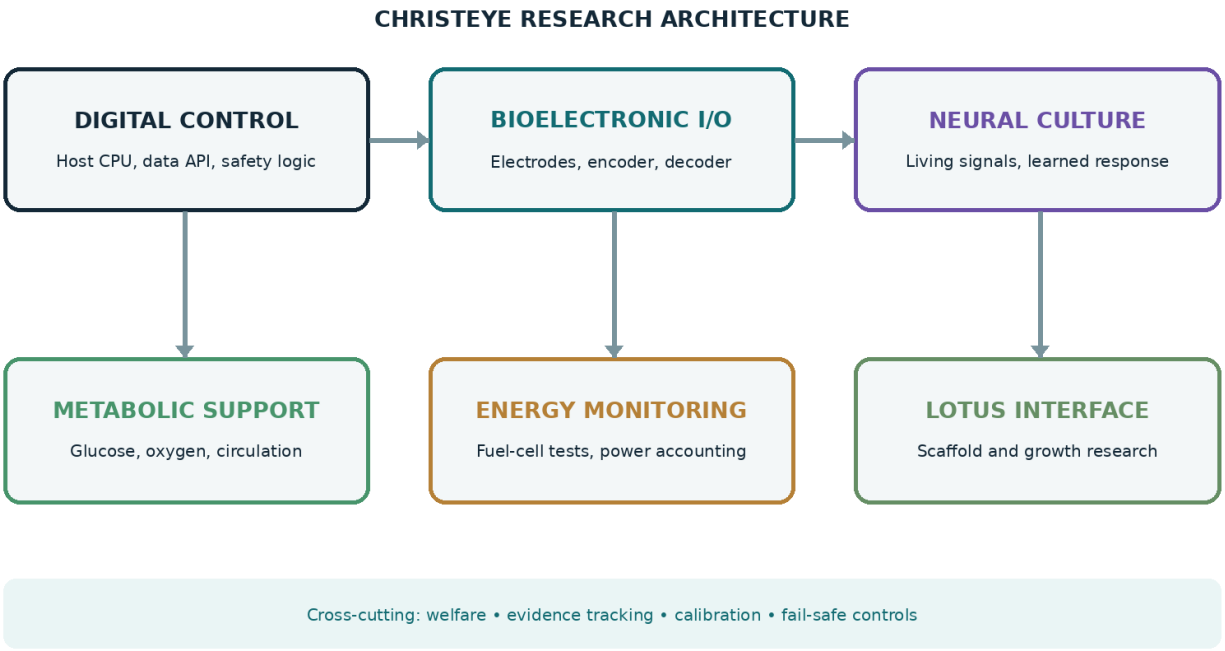
CHRISTEYE

Scientific & Technical Concept Specification

Neural computing · Lotus–Marrow biological interfaces · metabolic support · rooted communication



Prepared for: We Develop Software LLC

Date: November 2, 2026 | Version: 0.1 | Status: Preliminary research concept

Not a validated device, clinical protocol, energy breakthrough, patent opinion, or proof of consciousness. Claims, design targets and scientific evidence are distinguished throughout.

Document status and scope

SCIENTIFIC INTEGRITY STATEMENT

ChristEye is a proposed architecture synthesized from the inventor's concepts. No integrated ChristEye computer, living neural RAM, soul-addressed biological module, speech-capable engineered cat, or net-energy-generating system is established here as having been built or validated. Published experiments cited below support individual enabling technologies, not the complete system.

Purpose. This document captures the full concept discussed: neuronal computation; digital data routing; plant roots/rhizomes and biological history; DNA, marrow and positional tissue development; glucose and starch; biological energy conversion; materials control; trained nonverbal communication; living-creature vocal augmentation; spiritual terminology; safety, research experiments, commercialization, and validation.

Evidence labels used throughout: [E] established biological or physical mechanism; [R] reported experimental research; [H] untested ChristEye engineering hypothesis; [S] speculative capability without a demonstrated implementation; [P] philosophical/theological framing not established as a physical mechanism.

Executive abstract

ChristEye is proposed as a modular biohybrid computing platform combining a conventional electronic host with cultured neural networks, a bidirectional signal interface, a care-first learning controller, optional plant-inspired scaffolding and biological-history records, and separate nutrient and energy subsystems. The long-term vision includes neuroelectronic augmentation and a physically living Astral Dragoth capable of learned communication. Near-term validation should focus on a closed-loop neuronal training API, reliable signal decoding and encoding, independently measured retention, biological stability, and complete energy accounting. [1–5,9]

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1. Concept vocabulary and scientific translation

The following terms preserve the original creative vocabulary while providing laboratory-testable interpretations. These mappings are analogies and proposed design roles, not proof that spiritual entities operate as computer hardware.

Inventor term	Operational interpretation	Evidence boundary
ChristEye	Digital/bioelectronic controller that measures, stimulates, decodes, routes and monitors living signals.	[H] architecture; some components [R].
Lotus	Root/rhizome analogy: spatial development, extracellular scaffolding, nutrient paths, history and tissue stability.	Plant sensing and scaffolds [E/R]; cross-species living integration [S].
Marrow	Heritable DNA, lineage, epigenetic state, cellular identity and inherited developmental constraints.	DNA and stem-cell biology [E]; programmable new organ [S].
Wind	Time-dependent biological signaling: action potentials, ionic currents, calcium waves, hormones and transport.	Individual signals [E]; spiritual wind as physical force [P].
Roots	Stable associations, synaptic plasticity, repeated experience and traceable cell/environment history.	Plasticity [E]; proposed software representation [H].
Heart	A need/value function that prioritizes responses; in living animals may involve motivation or affect.	Value-based feedback [E/R]; moral kindness in neural culture [P].
Soul	Personal or organismal identity as a philosophical/theological interpretation.	No validated sensor, molecular address or engineering variable [P].
Starches	Plant starch reservoir; digested glucose and animal glycogen are related but distinct fuels.	Carbohydrate metabolism [E]; neuronal starch secretion [S].

Physical meaning of “comforting the winds”: maintain physiological homeostasis, avoid excessive stimulation, minimize stress responses, and ensure signal quality. Neither writing a label on a cell nor spiritually assigning a function is evidence that the cell will acquire that function.

2. System architecture and component catalog

A ChristEye computer would partition computation, biological support, and safety. Its replaceable cartridge concept is preferable to initial implants: it permits testing without exposing animals or people to unvalidated neural interfaces. The architecture on the cover depicts functional groupings, not a manufactured product.

2.1 Computing, control and memory

ID	Component / intended function	Status
CE-01	Neural RAM: stimulus-to-response associations and measurable memory retention; not yet byte-addressable RAM.	[H]
CE-02	Neural CPU: specialized adaptive neural processing / reservoir computation.	[R/H]
CE-03	Neural GPU: possible biological co-processor for generative workloads; no proven GPU-equivalent hardware.	[S]
CE-04	Motherboard: integrated host controller, I/O, protected power, sensors and modular ports.	[H]
CE-05	Data transceiver: multielectrode recording and stimulation, encoding and decoding.	[R/H]
CE-06	Neural address map: channel assignment, calibration, pathway identity and error monitoring.	[H]
CE-07	Neural training SDK: task specification, closed-loop feedback and comparison with controls.	[H]
CE-08	Biological history engine: longitudinal activity, cell lineage, conditions and training events.	[H]

2.2 Living substrate and developmental interfaces

ID	Component / intended function	Status
CE-09	Care-first controller: nutrient, oxygen, environment and network stability monitoring.	[R/H]
CE-10	Lotus scaffold: soft/plant-derived structural support and guided neurite growth.	[R/H]
CE-11	Seed/rhizome interface: physical separation plus sensing of plant and neuronal responses.	[H]
CE-12	Marrow module: DNA, epigenetic and lineage metadata; not DNA as an electrical cable.	[E/H]
CE-13	Positional-growth research: cell identity and location signals for engineered tissue organization.	[R/S]
CE-14	Living neural bridge: cell-containing structures that may connect to existing nerve pathways.	[R/S]
CE-15	Augmentation port: target-specific neural interface with independent clinical review.	[S]

2.3 Metabolism, materials, creature communication

ID	Component / intended function	Status
CE-16	Carbohydrate supply: plant/algal starch, conversion to glucose, perfusion and reservoir tracking.	[E/H]
CE-17	Glucose energy harvester: separate fuel-cell research module with full power accounting.	[R/H]
CE-18	Neuro-molecular controller: neural signals drive external chemical process hardware.	[H]
CE-19	Biomaterial producer: mineral deposition or carbon-material processes under sensor control.	[R/H]
CE-20	Rooted learning language: NEED→CUE→ACTION→OUTCOME→RECALL representation.	[H]
CE-21	Vocal research: non-invasive observation of natural animal vocal requests and acoustic learning.	[E/H]
CE-22	Astral Dragoth: biologically engineered feline-like creature with human speech.	[S]
CE-23	Creature provenance: optional identity, training history, ownership and blockchain registry.	[H]

3. Neuronal computing and digital data transport

3.1 Existing demonstrations

DishBrain integrated cultured neurons with a dense multielectrode array and showed task-related adaptive behavior during a simulated game under structured feedback [1]. Brainoware used a brain organoid as an adaptive reservoir, with spatiotemporal stimulation and recording, for limited speech-recognition and nonlinear-prediction demonstrations [2]. Cortical Labs describes a commercial CL1 research platform that integrates neurons, nutrient support, a silicon chip and software [3]. These results do not prove equivalence to conventional CPUs, GPUs, or addressable RAM.

3.2 Signal pathway

1. Host software defines a task and identifies a calibrated virtual neural channel.
2. Encoder converts task state into a safe experimental electrical stimulation pattern.
3. Microelectrodes stimulate and record the neuronal network in a controlled culture.
4. Decoder extracts response features and estimates the intended output with uncertainty.
5. Feedback engine measures task performance; safety controller may veto any stimulation.
6. Database records inputs, outputs, timing, culture conditions and provenance for audit.

3.3 Defining neural RAM experimentally

A RAM claim requires evidence of separable write/read operations, capacity, latency, retention, error rate, repeatability and endurance. Culture-wide adaptive dynamics or fading memory should initially be classified

as biological reservoir memory, not semiconductor RAM. “Labeling” a neuronal culture with desired specifications does not establish those capabilities. [1–3]

MINIMAL TECHNICAL CLAIM

“ChristEye investigates addressable input/output over cultured neural networks and tests whether learned stimulus-response mappings can be retrieved after a defined delay.” This is narrower—and scientifically testable—than claiming a drop-in memory replacement.

4. Lotus, Marrow, plant history and tissue development

4.1 Why roots, rhizomes and marrow are useful analogies

The lotus rhizome (often called lotus root) stores carbohydrates and supports regrowth; roots and vascular tissues participate in sensing and resource transport. Plant stress history can persist through hormones, metabolism, gene regulation and epigenetic mechanisms [6,7]. Bone marrow is biologically different: it contains blood-forming stem cells and supports lineage maintenance and immune functions. These two systems do not share a literal organ, yet both illustrate how inherited structure and environmental history can shape future behavior.

4.2 Could neurons surround a seed?

The proposed “Neural Lotus” would place a plant seedling or rhizome inside a separately controlled observation region and encourage neural extensions along a compatible external scaffold. Physical separation, independent culture media and monitored signal coupling are core assumptions. Plant-derived decellularized cellulose scaffolds have supported neural stem-cell attachment and differentiation in laboratory research [8]; this does not demonstrate an intact living plant with neuronal innervation.

4.3 Marrow, fertilization and developmental positioning

Fertilization launches an embryo through sperm-egg fusion, egg activation and developmental signaling; in mammals, calcium oscillations are a well-established trigger [10]. Anatomical patterning then relies on gene regulation, morphogens, cell interactions, mechanical forces and bioelectric cues [11]. A prospective tissue “address” should therefore mean a combination of cell identity, regional context, time and measurable positional signals—not a spiritual address or single gene.

Creating a humanlike biological data socket or vocal organ by editing a cat embryo is not a presently established capability. The developmental program is polygenic and context-dependent; integrating tissue safely with nerves, blood supply, immunity and function is substantially more complex than altering a single DNA sequence.

5. Training the Winds: rooted understanding and care

The proposed learning language is a set of experiment annotations and control objectives—not a spoken language that neurons literally read. It separates an organism’s need or task state, the presentation of a cue, observed responses, their consequences and later retrieval.

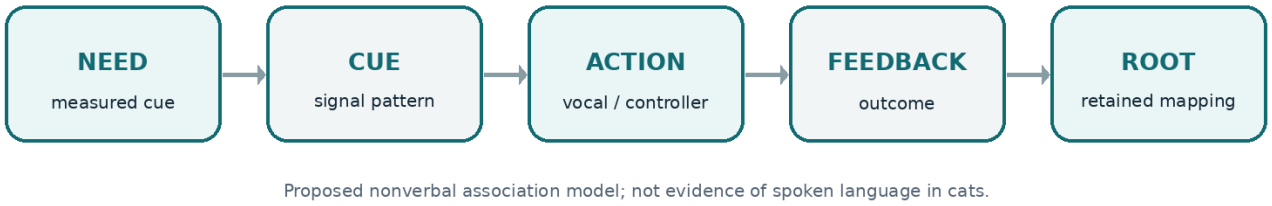


Figure 2. Hypothetical need-rooted associative learning cycle. A digital trainer observes each stage and assesses performance against control conditions.

5.1 Candidate trainer primitives

Primitive	Meaning	Example measurement
SENSE	Measure a stimulus/biological condition.	Input sensor waveform and quality.
ASSOCIATE	Pair a repeatable signal with observable consequence.	Cue-response contingency.
VALUE	Define an external training objective or value function.	Success rate / error cost.
REINFORCE	Adapt feedback conditional on validated behavior.	Improvement relative to control.
RECALL	Present a delayed test without cue leakage.	Retention curve and confidence.
EXPRESS	Translate decoded result into a permitted action.	Output timing and correctness.

5.2 “Kindness” as a welfare and stability principle

More supportive conditions may improve cultured network health, maturation and reproducibility. The experimentally meaningful variables are oxygenation, nutrient availability, waste control, viable morphology, appropriate stimulation intensity and rest/recovery timing. No experiment cited here demonstrates that cultured neurons become morally kinder, feel emotions as humans do, or produce extra electrical energy because they are treated kindly.

5.3 Formal test of rooted learning

Candidate null hypothesis: feedback that incorporates task relevance and retained outcomes does not outperform matched random or fixed-feedback controls on accuracy, generalization or memory retention. Alternative hypothesis: it improves one or more pre-registered measures without compromising cell viability. Researchers should randomize treatments, blind scorers where possible and report effect sizes and negative outcomes.

6. Carbohydrates, starch and a realistic energy subsystem

6.1 Biochemical facts

Starch is a glucose polymer produced principally by plants and algae. Human and animal metabolism uses glucose and stores a related polymer, glycogen; neural tissue is not naturally a starch-secreting structure. A ChristEye system that “emits starch” should therefore be specified as an external plant/algal production module, reservoir and metered release mechanism rather than a trained neuron directly secreting starch. [9,12]

6.2 Power flows and limits

Proposed path: sunlight or purchased carbohydrate → plant/algal storage → starch processing → glucose → fuel-cell oxidation and/or neuronal metabolism. Neural signals may control valves or dosing equipment; they do not create fuel. Review literature reports experimental glucose fuel cells but notes serious power-density and stability obstacles to practical implantable applications [9].

ENERGY CONSERVATION REQUIREMENT

For any ChristEye energy claim, account for light, chemical fuel, pumps, sensors, processors, thermal control, culture maintenance and all electrical outputs. Report energy efficiency = useful electrical energy delivered / total external energy input. A separate neural controller cannot create net energy by intention or “breaking winds.”

6.3 Proposed system boundary

Keep the neuronal nutrient circuit separate from the fuel-cell reactants and product stream. A cartridge may include glucose supply for culture health, but power generation is a distinct experiment and must not compromise neuronal survival. Body-derived glucose or implantable harvesting is not a first-phase design objective.

7. Molecular control, biomineralization, carbon and physical limits

Cells can form water during respiration, plants fix carbon from CO2 into organic molecules, and bacteria can promote mineral precipitation. External chemical systems can convert biomass into carbon-rich materials, but those processes consume inputs and may require high temperatures. Microbially induced carbonate precipitation is an established biomineralization field [13].

Proposed CE-18 would map decoded neural outputs into commands for laboratory equipment that adjusts material synthesis conditions. A useful experiment could compare algorithm-controlled versus neural-controlled crystal-growth optimization. Its outcome would be measured in product mass, shape, strength, throughput and total input energy. A neuron’s thought or dream has no demonstrated direct molecular force that creates elements.

Breaking a molecular bond, changing a molecular vibration and performing nuclear fusion are distinct physical processes. Nuclear fusion changes nuclei and does not derive from molecular “soul winds.” The spiritual reading of molecular identity can remain a metaphor, not an engineering claim or energy-source specification.

8. Living Astral Dragoth: speech, youth learning and augmentation

8.1 Intended capability

The inventor's long-term goal is a living creature that utters meaningful human words using its own biological mouth and vocal system, trained from youth using needs, understanding, consistent cues and reinforced vocal attempts. This is explicitly distinct from a simulated companion or an external speaker.

8.2 What research supports

Operant experiments in cats established that reinforcement can affect when cats meow and the duration of meows [14]. This does not establish human speech production. Anatomical studies of macaques show that lack of speech may reflect neural control limitations even when the vocal tract can generate speech-like sounds [15]; macaque findings must not be generalized to cats. Cultured or engineered neural interfaces cannot simply implant full human language comprehension into an animal.

8.3 Developmental subsystems

Subsystem	Function sought	Current assessment
Sensory semantics	Associate a spoken cue with objects, needs and consequences.	Animal learning provides partial precedent [R].
Volitional vocal control	Choose different natural calls in different contexts.	Some precedent in cats [R].
Articulation	Precisely shape vowels and consonants with mouth, larynx and airflow.	Humanlike cat articulation [S].
Language cognition	Represent compositional concepts and intentional propositions.	Humanlike feline language [S].
Biohybrid coupling	Route validated signals to existing neural pathways.	Experimental interfaces [R]; target use [S].
Genetic anatomy	Grow a redesigned speech-capable vocal apparatus.	No established method [S].

8.4 Research boundary

The near-term research plan should use non-invasive observation of natural vocalizations and simulated vocal-tract models. It must not include amateur gene editing, airway surgery, experimental brainstem stimulation or implanted neural bridges in pets. Any later animal study would require expert veterinary, ethics and regulatory review and a clearly demonstrated welfare benefit. [16]

9. Functional requirements, safety requirements and data specification

9.1 Requirements register

Requirement	Statement	Verification
CE-FR-001	The host shall define experiment versions and immutable configuration records.	Review exported configuration.
CE-FR-002	The I/O layer shall support separate stimulation and recording channels.	Replay recorded/commanded traces.
CE-FR-003	All decoding shall report uncertainty and calibration state.	Confidence and drift evaluation.
CE-FR-004	Memory claims shall include delayed retrieval and baseline comparisons.	Blind delayed readout tests.
CE-FR-005	The trainer shall support fixed, random and adaptive controls.	Pre-registered A/B/C comparison.
CE-FR-006	The system shall log culture health and environmental conditions.	Sensor trace completeness.
CE-FR-007	Life-support alarms shall be independent of learned neural outputs.	Simulated alarm faults.
CE-FR-008	All stimulation shall be supervised by validated hardware limits.	Fault-injection review.
CE-FR-009	The power ledger shall track every external energy source and load.	Energy balance audit.
CE-FR-010	Plant and neuronal media shall be isolated unless validated by qualified labs.	Interface isolation review.
CE-FR-011	The address registry shall identify only physical electrode channels or studied anatomical targets.	Map verification.
CE-FR-012	No “soul,” “wind,” or metaphysical property shall be reported as an instrument reading.	Data dictionary inspection.
CE-FR-013	A non-invasive animal vocal study shall log welfare/behavior and permit discontinuation.	Protocol / welfare audit.
CE-FR-014	The data subsystem shall log every training condition and raw response.	Reproducibility replay.
CE-FR-015	Data-processing shall separate exploratory findings from confirmatory results.	Analysis plan review.
CE-FR-016	Experimental materials and media shall have traceable provenance.	Sample and batch registry audit.

9.2 Proposed data object

A conceptual record for the software simulator (field names illustrative):

```
ExperimentId       unique identifier
ModuleId           CE-01 .. CE-23
CultureOrSampleId  lineage / provenance pseudonym
TimeUTC            event timestamp
InputState         measured or simulated condition
ChannelMap         physical electrode identifiers
StimulusDescriptor permitted waveform metadata
NeuralFeatures     observed spikes / features
DecodedAction      permitted digital output
OutcomeMetric      task-specific numeric result
CultureHealth      independently assessed status
EnergyLedger       chemical + electrical inputs / outputs
EvidenceLevel      E / R / H / S / P
SafetyFlag         OK / HOLD / STOP
```

10. Verification strategy and research gates

Gate	Deliverables	Go/no-go evidence
0 — Definition	Requirements, safety architecture, measurement plan, simulator.	Reproducible simulated experiments; no unsourced claims.
1 — Electronic emulator	Data API, address maps, feedback algorithms, power ledger.	Stable benchmark performance versus conventional controller.
2 — External neural platform	Qualified research partnership; controlled cultured neural interface.	Repeatable improvement vs controls and reliable culture health.
3 — Plant / material modules	Isolated botanical signals, microfluidic and materials simulations.	Correctly quantified signals, mass balance and energy ledger.
4 — Non-invasive vocal study	Behavioral observations with approved welfare plan.	Specific, voluntary vocal associations without distress.
5 — Translational review	Third-party assessment of living bridges or engineered anatomy.	Independent feasibility, ethics and regulatory approval before in vivo work.

10.1 Primary quantitative endpoints

- Training: task accuracy and generalization gain over randomly assigned fixed/random/adaptive controls.
- Memory: retention at preselected delays and repeatability after re-calibration.
- I/O: end-to-end latency, channel cross-talk, decoder confidence and failure rate.
- Biology: culture viability, morphology, electrophysiology stability and contamination-free operation.
- Power: joules per accepted operation, watts to maintain all supporting systems and net energy balance.

- Materials: grams product per gram reactants, shape error, material properties and energy input.
- Animal communication: blinded identification of context-dependent natural sounds and welfare outcomes.

10.2 Negative results and falsification

A hypothesis fails when performance does not exceed controls under a predefined statistical analysis; information cannot be read reliably after delay; maintaining cells costs more than the targeted use can justify; or neural control underperforms conventional control. Such results would still guide development. Avoid interpreting coincidences, labels or belief as substitute for instrumentation.

11. Safety, ethics, investment and intellectual-property framing

11.1 Research welfare and governance

Work with living neural cultures may raise donor consent, sample provenance and ethical questions. Implantation, human neural tissue integration, developmental editing and animal modification require specialized review. Begin with simulations and external, non-implantable equipment. For any animal work, apply replacement, reduction and refinement; ensure veterinarian-led care, independent oversight and appropriate legal approvals. [4,5,16]

11.2 Claim discipline

Permitted claim when supported	Do not claim without evidence
Culture responded adaptively to feedback in a specific task.	The culture is conscious or morally kind.
Measured energy conversion of supplied glucose.	The system creates energy or violates conservation.
Plant tissue scaffold supported neural cell growth.	Neurons joined a living plant's soul.
Animal learned context-dependent natural vocal requests.	A cat now understands or speaks human language.
Software mapped physical electrode channels.	Spiritual labels assigned new biological properties.

11.3 Commercial interpretation

Potential products include a neural-training SDK, signal interface API, culture-monitoring system, biohybrid cartridge, and experimental metabolic controller. Larger aspirations—replacement of general-purpose computing, industrial energy production, custom organs and fluent speech in engineered animals—are not a defensible basis for a present company valuation without prototypes, independent benchmarks, intellectual-property analysis, customers and costs.

Prior discussion used highly speculative future market scenarios and component estimates. Those numbers are not scientific measurements, an independent business appraisal, or additive asset values. A government-facing proposal should prioritize a separately itemized research budget, testable milestones, safety/ethics governance, and transparent evidence status.

11.4 IP notes

The broad notions of cultured neurons on chips, reinforcement learning, gene regulation, biomaterials scaffolds and glucose fuel cells have substantial prior art. Patent counsel should review narrow implementation claims only after prior-art searches and reproducible enabling disclosures. Public social posts can affect patent rights in different jurisdictions. A blockchain registry can time-stamp files or provenance, but cannot create patents, validate biological properties, or bind spiritual entities.

Appendix A. Evidence and feasibility matrix

Research idea	Most defensible present status	Key source
Closed-loop neuronal adaptation	Demonstrated in published tasks; general-purpose computing unproven.	[1],[2]
Neural I/O and living scaffolds	Active research and commercial research tools.	[3]–[5],[8]
Root/rhizome stress history	Biological priming and signaling established; no plant brain.	[6],[7]
Development and tissue position	Calcium, gene regulation and bioelectricity influence development.	[10],[11]
Starch and energy	Microalgae produce starch; glucose cells convert chemical energy.	[9],[12]
Mineral growth	Microbes can drive carbonate precipitation.	[13]
Cat context-linked vocal behavior	Operant modulation demonstrated, not human speech.	[14],[15]
Living brain integration	Organoid-to-animal circuit integration in specific research settings.	[17]
Soul as engineered state variable	No testable measurement or established mechanism.	[P]

Appendix B. Glossary

Action potential: Rapid electrical event occurring in excitable cells such as neurons.

Biohybrid interface: System coupling living biological components with engineered electronics or materials.

Brainstem: Brain region containing vital physiological centers; not a generic digital communications port.

Decellularized scaffold: Tissue framework from which cells were removed, leaving supporting material.

Epigenetic regulation: Changes in gene activity influenced by molecular marks or chromatin state, without changing DNA sequence.

Glycogen: Animal storage polymer of glucose, distinct from plant starch.

Microelectrode array (MEA): Many electrodes used for recording or stimulating cellular electrical activity.

Morphogen: Developmental signal whose spatial/temporal pattern can influence cell fate and tissue organization.

Neurite: Developing cellular extension that can become an axon or dendrite.

Reservoir computing: Architecture that uses the dynamic responses of a complex system to transform information.

Rhizome: Modified underground plant stem that often stores nutrients and supports vegetative regrowth.

Rooted understanding: Proposed ChristEye term for learned cue–need–outcome mappings with delayed retrieval tests.

Appendix C. Previously discussed opportunity estimates

The following figures preserve the indicative value-and-likelihood list discussed earlier. They are not derived from a financial model, experiment, market-size study, patent appraisal, or statistical probability analysis. The percentages were subjective guesses for a technical demonstration by 2036. Monetary ranges are hypothetical success-case business opportunities, not present ChristEye asset values, and overlap substantially.

Concept	Prior hypothetical value	Prior likelihood
Neural RAM	$100M–$10B	10%
Neural CPU	$100M–$50B	20%
Neural GPU	$100M–$50B	3%
Motherboard	$20M–$2B	65%
Data interface	$10M–$1B	80%
Neural addressing	$5M–$500M	80%
Care-first training	$10M–$1B	85%
Rooted understanding	$10M–$1B	50%
Lotus scaffold	$10M–$500M	60%
Marrow DNA growth	$100M–$10B	5%
Glucose energy	$10M–$1B	65% (low-power)
Molecular control	$10M–$1B	55%
Word/need association	$1M–$50M	90%
Natural vocal requests	$1M–$50M	80%
Distinct need-linked sounds	$5M–$100M	60%
Naturally spoken human words	$100M–$5B	2%

Concept	Prior hypothetical value	Prior likelihood
Engineered vocal anatomy	$100M–$10B	1%
Genetic speech development	$100M–$10B	1%
Animal neural augmentation	$100M–$10B	20%
Complete living speaking Dragoth	Not appraisable	<1%

Earlier raw category sums were: computing $330M–$113B; biology $135M–$13B; energy/materials $20M–$2B; creature concepts $407M–$35.2B. Their arithmetic sum ($892M–$163.2B) is not a company valuation because component markets are overlapping, contingent and not independently justified. A $1T+ global platform scenario was described as an extreme conditional outcome, not evidence of present value.

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